

## PRACTICAL WEEK-2

### TASK1

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laasya@laasya:~$ mkdir practical_week2
laasya@laasya:~$ cd practical_week2
laasya@laasya:~/practical_week2$ pwd
/home/laasya/practical_week2
laasya@laasya:~/practical_week2$ nano command_executor.c
laasya@laasya:~/practical_week2$ ls
command_executor.c
laasya@laasya:~/practical_week2$ gcc command_executor.c -o command_executor
laasya@laasya:~/practical_week2$ ls
command_executor  command_executor.c
laasya@laasya:~/practical_week2$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 563
Child PID  : 566

Child Process
Parent PID : 563
Child PID  : 566

Executing command: ls
command_executor  command_executor.c

Child process completed.
laasya@laasya:~/practical_week2$
```

```
laasya@laasya:~/practical_week2$ uname -a
Linux laasya 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
laasya@laasya:~/practical_week2$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:              39 bits physical, 48 bits virtual
  Byte Order:                 Little Endian
CPU(s):                       12
  On-line CPU(s) list:       0-11
Vendor ID:                    GenuineIntel
  Model name:                 13th Gen Intel(R) Core(TM) i5-1334U
    CPU family:               6
    Model:                    186
    Thread(s) per core:       2
    Core(s) per socket:       6
    Socket(s):                1
    Stepping:                 3
    BogomIPS:                 4991.99
  Flags:                      fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:             VT-x
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Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
  L1d:                   288 KiB (6 instances)
  L1i:                   192 KiB (6 instances)
  L2:                    7.5 MiB (6 instances)
  L3:                   12 MiB (1 instance)
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:                Mitigation; Enhanced IBRS
  Spec rstack overflow:    Not affected
  Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DISABLE
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
laasya@laasya:~/practical_week2$ ps
  PID TTY          TIME CMD
  307 pts/0    00:00:00 bash
  588 pts/0    00:00:00 ps
laasya@laasya:~/practical_week2$ px aux
Command 'px' not found, but can be installed with:
```

```

laasya@laasya:~/practical_week2$ top
top - 05:00:21 up 12 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni, 99.9 id,   0.1 wa,   0.0 hi,   0.0 si,   0.
MiB Mem :  7762.6 total,  7217.2 free,   515.5 used,   180.0 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7247.1 avail Mem

```

| PID | USER     | PR | NI | VIRT    | RES   | SHR   | S | %CPU | %MEM | TIME+   |
|-----|----------|----|----|---------|-------|-------|---|------|------|---------|
| 1   | root     | 20 | 0  | 21652   | 13124 | 9796  | S | 0.0  | 0.2  | 0:00.57 |
| 2   | root     | 20 | 0  | 3180    | 2208  | 2072  | S | 0.0  | 0.0  | 0:00.01 |
| 7   | root     | 20 | 0  | 3196    | 2128  | 2008  | S | 0.0  | 0.0  | 0:00.00 |
| 53  | root     | 19 | -1 | 50428   | 16232 | 15136 | S | 0.0  | 0.2  | 0:00.16 |
| 103 | root     | 20 | 0  | 25136   | 6452  | 5132  | S | 0.0  | 0.1  | 0:00.31 |
| 112 | systemd+ | 20 | 0  | 21460   | 13192 | 10936 | S | 0.0  | 0.2  | 0:00.09 |
| 113 | systemd+ | 20 | 0  | 91028   | 7988  | 7024  | S | 0.0  | 0.1  | 0:00.06 |
| 177 | root     | 20 | 0  | 4244    | 2680  | 2432  | S | 0.0  | 0.0  | 0:00.01 |
| 178 | message+ | 20 | 0  | 9596    | 5344  | 4660  | S | 0.0  | 0.1  | 0:00.04 |
| 191 | root     | 20 | 0  | 17968   | 8840  | 7796  | S | 0.0  | 0.1  | 0:00.08 |
| 194 | root     | 20 | 0  | 1755848 | 13124 | 10888 | S | 0.0  | 0.2  | 0:00.11 |
| 214 | syslog   | 20 | 0  | 222516  | 5956  | 4552  | S | 0.0  | 0.1  | 0:00.07 |
| 216 | root     | 20 | 0  | 3124    | 1972  | 1836  | S | 0.0  | 0.0  | 0:00.00 |
| 231 | root     | 20 | 0  | 107020  | 23192 | 13728 | S | 0.0  | 0.3  | 0:00.09 |
| 302 | root     | 20 | 0  | 3188    | 1112  | 980   | S | 0.0  | 0.0  | 0:00.00 |
| 306 | root     | 20 | 0  | 3204    | 1248  | 1108  | S | 0.0  | 0.0  | 0:00.06 |
| 307 | laasya   | 20 | 0  | 6080    | 5396  | 3676  | S | 0.0  | 0.1  | 0:00.08 |
| 309 | root     | 20 | 0  | 6700    | 4640  | 3860  | S | 0.0  | 0.1  | 0:00.01 |
| 353 | laasya   | 20 | 0  | 20316   | 11644 | 9516  | S | 0.0  | 0.1  | 0:00.09 |
| 354 | laasya   | 20 | 0  | 21156   | 3580  | 1852  | S | 0.0  | 0.0  | 0:00.00 |
| 379 | laasya   | 20 | 0  | 6080    | 5336  | 3676  | S | 0.0  | 0.1  | 0:00.02 |
| 601 | laasya   | 20 | 0  | 9300    | 5680  | 3496  | R | 0.0  | 0.1  | 0:00.01 |